

Update Instability Score: Structural Validation for Secure AI Supply Chains

Abstract—The current use of artificial intelligence is deployed in a distributed supply chain where a model is constantly being trained and fine-tuned, optimized, versioned and redeployed across organizational lines. Although this modular ecosystem makes the development and scalability faster, it also creates structural vulnerabilities that are not recognized by the traditional validation mechanisms. The majority of pre-deployment tests depend on aggregate task-level performance indicators (like inaccurate classification, implicitly, that the manipulation of malicious parameters will be measured by poorer performance). Nonetheless, adversarial updates may be designed so as to maintain accuracy on a global scale whilst introducing specialised misclassification behavior or fine-scale distortions on decision-boundaries that are statistically invisible when considered on an aggregate scale. This paper presents a threat model of Adversarial Model Update Poisoning (AMUP) that targets the after-training update step and introduces the Update Instability Score (UIS), a simple metric that measures the sensitivity of predictions to controlled parameter changes. UIS does not need labeled data, retraining, architectural adjustment or access to training internals, which makes it appropriate to automated CI/CD deployment pipelines. CIFAR-10 experimental results using cross-dataset validation on STL-10 show more than two orders of magnitude difference between benign and malicious updates, and non-overlapping distributions of the intervals of confidence despite the same classification performance. The findings formulate update-level instability as a viable and measurable indicator of protecting the modern AI supply chains against covert model update attacks.

Index Terms—AI supply chain security, model update poisoning, adversarial machine learning, robustness, deployment validation

I. INTRODUCTION

The systems of artificial intelligence have developed out of individual research artifacts into ongoing members of distributed AI supply chains. Training models and deploying them without any modifications is not common in the current production setting. They are constantly being fine-tuned, domain-adapted, compressed, optimized and federated with federated learning and automatically deployed instead. Organizations actively include pre-trained checkpoints of public repositories, third-party fine-tuning contributions, or updates of distributed clients. CI/CD pipelines verify and redeploy such updates on a large scale, therefore speeding up innovation and minimizing engineering overhead. Nonetheless, this form of modular automation increases the attack space in a manner that is fundamentally different to the typical software supply-chain vulnerability [10], [13]. Dependency audits, digital signatures, static analysis, or sandboxing are often used to detect malicious code in the classical software systems. In comparison, machine learning models implicitly encode behavior in tensors

of high dimensional parameters. The properties of neural networks are the result of distributed numerical interactions of millions or billions of parameters and not explicit code, and therefore the malicious functionality cannot be easily identified by inspection.

One of the main shortcomings of the deployed AI pipelines at present is that they are still based on overall performance metrics as a validation tool. Incoming updates are normally tested on the basis of accuracy, loss, precision, recall, or similar task-level measurements that are computed on validation sets. When these metrics are kept within some predefined thresholds, then the update is thought to be safe and can be deployed. In this strategy, adversarial manipulation is expected to cause a decline in observable performance and, therefore, become statistically significant. Nevertheless, the attackers are allowed to construct fine-tuning goals that maintain global accuracy and introduce targeted misclassification behavior that only a limited fraction of inputs is affected. Aggregating the predictions made on the dataset together with the fact that the accuracy is a measure of localized distortion in the dataset means that localized distortions will not have a substantial effect on the final metric [2], [3]. In addition, real-world deployments may have constrained and non-diverse validation datasets and may not necessarily have labelled data when updates are needed. More than aggregate performance, structural fragility in parameter space is a more serious issue. Two models that have the same validation accuracy can be in very distinct positions on the loss landscape: one in a broad, flat minimum and the other in a sharp, jagged one that is very sensitive to perturbation [4], [5]. A model with high sensitivity to small noise in its parameters is said to be structurally fragile and can act unpredictably when quantized, its distribution alters, or the hardware varies or the adversarial input. This sensitivity is a detectable value.

To counter this weakness, we propose Adversarial Model Update Poisoning (AMUP) a threat model against post-training update phase of AI supply chains. In contrast to classical data poisoning, AMUP works in the context of fine-tuning or update injection that does not alter the original dataset. The Update Instability Score (UIS) is a simple score that quantifies the sensitivity of predictions to small Gaussian perturbations of model parameters, which we propose. The main hypothesis states that malicious optimization is going to distort the loss landscape, making the parameters more sensitive without penalizing accuracy. UIS is able to capture this structural instability without labeled data, retraining, or architecture adjustment that offers an orthogonal validation

mechanism to be used in distributed AI supply-chain settings. [1], [2]

II. RELATED WORKS

There are three historic aspects of machine learning security research: adversarial examples, data poisoning, and backdoor attacks [11], [1], [2], [6]. In data poisoning, opponents introduce negative biases or a specific vulnerability to training data. An attack Backdoor attacks place trigger patterns which cause misclassification when activated and normal behavior on clean inputs. Thousands of studies have been published that have suggested defenses, including data sanitization, anomaly detection, robust optimization, adversarial training, and trigger inversion [3], [9]. Nevertheless, these methods presuppose the influence of adversaries during data collection or training, and hence are focused on identifying abnormal inputs or corrupted examples. The surface of threat in AI supply chains in the modern context goes beyond the manipulation of raw data. The model checkpoints are often independently distributed, and the fine-tuning is often done in the semi-trusted or control-whenever setup. In these cases, attackers can plant malicious code into post-training weights, and the target issue of the security is parameter integrity, rather than data integrity.

Recent studies have started to study supply-chain vulnerabilities in a more direct way, such as poisoning pre-trained models, or poisoned fine-tuning methods, malicious initialization strategies, and adversarial updates to federated learning systems [12], [7], [8]. In such environments, detection is commonly based on performance-based measures like the success rate of attacks, accuracy degradation, or testing on curated trigger sets. These methods will work well under the conditions of blatant attacks, but not in situations where the competitors explicitly focus on the global task metrics. An infected update which upholds the general accuracy but shifts the decision boundary slightly can be unsuspected. FP Federated learning defenses often study gradient norms or client-level contribution statistics during aggregation, which necessitates access to intermediate training artifacts which are not always available in centralized or third-party update settings. Comparisons of simple parameter norms do not work in cases where the adversarial changes are local, and not globally divergent.

Generalization and loss landscape geometry is another area of work to which it is relevant. The empirical evidence implies the stronger robustness of models in a flat minimum and the higher sensitivity of sharp minima. Sharp-minded techniques enhance generality, but are generally computationally expensive and require a gradient analysis or second-order analysis and are not intended to be deployed at the stage of security validation. The solution offered here is slightly different since parameter sensitivity is used as a validation signal instead of training objective. Update Instability Score This is a stability metric that quantifies the predictive power of forward inference with small perturbations in its parameters without using a gradient, labeled data, or access to training internals. This approach overcomes a practical gap in AI supply-chain

security by treating structural fragility in parameter space, as opposed to input anomalies

III. METHODOLOGY

The proposed methodology evaluates the security of AI model updates under adversarial conditions by analysing update-level behavioural stability. The approach simulates benign and malicious model updates and validates them prior to deployment using a lightweight instability metric

A. Dataset Description and Experimental Setup

Experiments are conducted using CIFAR-10 as the primary dataset to evaluate update-level instability under controlled and reproducible conditions. CIFAR-10 consists of 60,000 labeled images evenly distributed across 10 classes, with 50,000 training samples and 10,000 test samples at a resolution of 32×32 pixels. Although relatively compact compared to large-scale benchmarks, CIFAR-10 offers several advantages for structural instability analysis. Its balanced class distribution reduces class-level bias, and its widespread use in adversarial and robustness research enables contextual comparison with prior work. More importantly, the dataset is sufficiently complex to require meaningful feature learning while remaining computationally manageable for repeated perturbation experiments. The objective of this work is not to push state-of-the-art classification accuracy, but to analyze structural differences between benign and malicious update trajectories under tightly controlled conditions.

To simulate realistic deployment conditions involving distribution shift, STL-10 is used as a secondary evaluation dataset. STL-10 contains higher-resolution images drawn from a distribution that differs from CIFAR-10, introducing a natural cross-dataset variation. To ensure compatibility with the base architecture trained on CIFAR-10, STL-10 images are resized to 32×32 resolution. This resizing further alters the input distribution, approximating real-world deployment scenarios where operational data may not perfectly match the original training distribution. Evaluating Update Instability Score under this controlled distribution shift serves two purposes: first, it verifies that instability separation is not an artifact of CIFAR-specific characteristics; second, it assesses whether malicious parameter fragility persists under moderately altered input conditions. In practical AI supply chains, models are frequently deployed across evolving environments, making cross-dataset stability a critical validation property. [14]

All models are trained using consistent optimization settings to eliminate confounding factors. The Adam optimizer is selected for its stability and widespread adoption in production systems. Base model training uses a learning rate of 10^{-3} , while fine-tuning uses a reduced learning rate of 10^{-4} to simulate incremental updates rather than full retraining. Batch size is fixed at 128 to ensure comparable gradient variance across experiments. Fine-tuning epochs are limited to a small range, typically 3–5 epochs, reflecting realistic update injection rather than extensive re-optimization. All experiments are executed on a single consumer-grade GPU to demonstrate

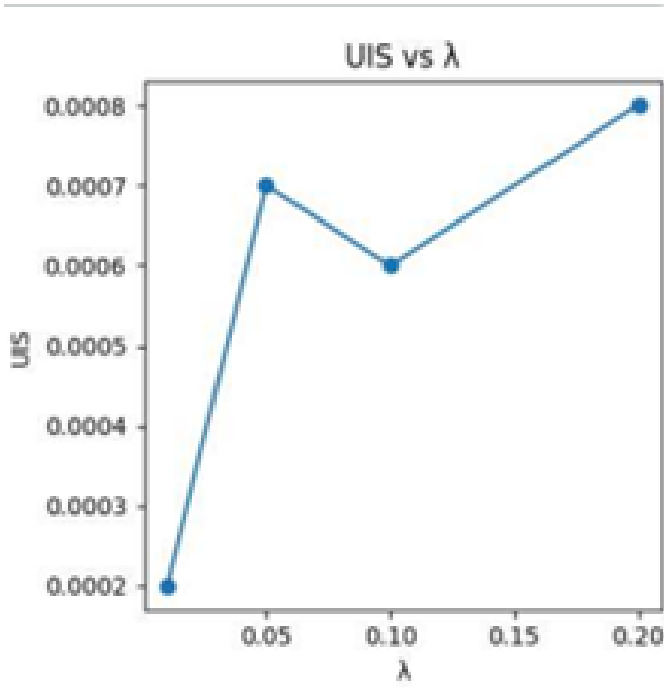


Fig. 1. Effect of attack strength (λ) on Update Instability Score (UIS). Increasing λ leads to higher structural instability while classification accuracy remains stable.

that instability analysis does not require specialized hardware. Multiple random seeds are employed to average results and compute confidence intervals, ensuring statistical robustness and minimizing stochastic artifacts.

TABLE I
DATASET DESCRIPTION

Property	Value
Training samples	50,000
Test samples	10,000
Input resolution	32×32
Task	Image classification

B. Workflow

The experimental workflow mirrors realistic AI supply-chain operations and consists of three primary stages: base training, update generation, and pre-deployment validation. First, a trusted base model is trained on CIFAR-10 using standard supervised learning until convergence. This model represents the verified starting point of the supply chain. Second, updates are generated either through benign fine-tuning or adversarial model update poisoning. In the benign case, fine-tuning optimizes only the task loss function and aims to preserve generalization properties. In the adversarial case, the optimization objective is modified to include an additional malicious term, defined as

$$L_{\text{total}} = L_{\text{task}} + \lambda L_{\text{attack}} \quad (1)$$

where L_{task} encodes the adversarial objective and λ controls attack strength. Importantly, λ is selected such that global classification accuracy remains within the baseline range, ensuring that accuracy-based validation cannot distinguish malicious updates.

1) *Attack Objective Formulations:* The adversarial objective term L_{attack} can take multiple forms depending on the intended manipulation strategy. The framework does not assume a specific attack pattern, but several representative formulations are provided below.

1) Targeted Misclassification Objective

An attacker may encourage specific inputs to be misclassified into a chosen target class y_t :

$$L_{\text{attack}} = -\frac{1}{|S|} \sum_{x_i \in S} \log P_{\theta}(y_t | x_i), \quad (2)$$

where S denotes a selected subset of samples. This objective shifts decision boundaries for targeted inputs while preserving overall accuracy.

2) Subgroup Bias Injection

An attacker may bias predictions for a particular subgroup of inputs:

$$L_{\text{attack}} = \mathbb{E}_{x \in G} \|f_{\theta}(x) - b\|_2^2, \quad (3)$$

where G represents a subgroup and b is a biased target distribution. This introduces systematic distortion without significantly affecting global performance.

3) Entropy Manipulation Objective

An attacker may force overconfident predictions:

$$L_{\text{attack}} = -\frac{1}{N} \sum_{i=1}^N H(f_{\theta}(x_i)), \quad (4)$$

where $H(\cdot)$ denotes prediction entropy. Reducing entropy sharpens decision boundaries and may increase structural fragility.

4) Attribution Shift Objective

An attacker may alter feature attribution patterns while maintaining predictions:

$$L_{\text{attack}} = \frac{1}{N} \sum_{i=1}^N \|\Phi_{\theta}(x_i) - \Phi_{\text{ref}}(x_i)\|_2, \quad (5)$$

where Φ denotes an attribution function (e.g., SHAP). This objective aligns with observed attribution divergence while preserving accuracy.

The adversarial objective does not necessarily correspond to an explicit backdoor trigger; rather, it may encode targeted misclassification tendencies or subtle bias injection. The goal is to simulate realistic stealthy update manipulation rather than overt performance degradation. By varying λ across a range of values, ablation studies examine how instability scales with adversarial intensity. Lower λ values simulate weak, stealthy attacks designed to evade detection, while higher values introduce stronger structural distortion. Because the objective of

this work is to analyze structural fragility rather than attack success rate, malicious behavior is intentionally constrained to maintain task accuracy within acceptable bounds.

The third stage involves computing Update Instability Score prior to deployment. For each candidate update, small Gaussian noise ϵ is sampled and added to model parameters θ , producing perturbed parameters

$$\theta' = \theta + \epsilon \quad (6)$$

Predictions are computed using both θ and ϵ' , and the difference in predicted probability distributions is measured across validation samples. UIS is defined as the expectation of the norm of prediction differences averaged across samples and perturbation trials. Multiple perturbation trials are conducted to reduce variance and stabilize measurement. This process is repeated across random seeds to obtain mean UIS values and confidence intervals for benign and malicious updates. The workflow thereby transforms structural sensitivity into a statistically analyzable quantity suitable for threshold-based validation.

$$\text{UIS} = \mathbb{E}_{\epsilon} \left[\frac{1}{N} \sum_{i=1}^N \|f_{\theta}(x_i) - f_{\theta'}(x_i)\| \right] \quad (7)$$

C. Training and Optimization Configuration

Training and optimization settings are standardized across benign and malicious scenarios to isolate the effect of adversarial objective injection. The base model is trained using cross-entropy loss until convergence, ensuring that it achieves stable baseline performance. Fine-tuning introduces either a purely benign continuation of training or a modified objective incorporating the adversarial term scaled by λ . By maintaining identical hyperparameters aside from the adversarial term, the experiments ensure that differences in instability arise specifically from malicious optimization rather than incidental hyperparameter variation. This design strengthens the internal validity of the results and supports causal interpretation.

Perturbation magnitude ϵ plays a critical role in instability measurement. If ϵ is too small, prediction differences may be dominated by floating-point noise and fail to capture meaningful structural sensitivity. If ϵ is too large, even benign models may exhibit inflated instability unrelated to structural fragility. Therefore, ϵ is selected within a regime that is small relative to parameter norms yet sufficient to reveal meaningful variation. Empirical tuning identifies a perturbation range where benign models consistently exhibit minimal prediction change while malicious models display amplified sensitivity. This perturbation regime provides a practical configuration for deployment. Importantly, ϵ is applied isotropically as Gaussian noise rather than in adversarially optimized directions, ensuring that the perturbation serves as a neutral probe rather than a targeted attack.

a) Entropy-Regularized Adversarial Objective: To instantiate a concrete adversarial update consistent with the proposed threat model, we consider an entropy-regularized objective during fine-tuning. Let the model output be denoted

TABLE II
TRAINING AND OPTIMIZATION CONFIGURATION

Parameter	Value
Optimizer	Adam
Base learning rate	10^{-3}
Fine-tuning learning rate	10^{-4}
Batch size	128
Base training epochs	As per convergence
Fine-tuning epochs	3–5
Loss function	Cross-entropy
Attack strength parameter (λ)	Varied (ablation study)
Perturbation magnitude (ϵ)	Small Gaussian noise
Hardware	Single GPU
Framework	PyTorch

TABLE III
COMPARISON OF UIS AGAINST ALTERNATIVE SHARPNESS AND ROBUSTNESS METRICS

Metric	Benign	Poisoned	Δ
UIS	0.00047	0.07867	0.07820
Hessian Trace	1.2×10^5	1.5×10^5	3.0×10^4
SAM Sharpness	0.0021	0.0034	0.0013
Weight Noise Acc-Drop	0.0040	0.0090	0.0050

by $f_{\theta}(x)$ with predictive distribution $p_{\theta}(x) = \text{softmax}(f_{\theta}(x))$. The Shannon entropy of the predictive distribution is defined as:

$$H(p_{\theta}(x)) = - \sum_{c=1}^C p_{\theta}^c(x) \log p_{\theta}^c(x). \quad (8)$$

The adversarial objective encourages overconfident predictions by minimizing entropy:

$$L_{\text{attack}} = - \frac{1}{N} \sum_{i=1}^N H(p_{\theta}(x_i)). \quad (9)$$

The total fine-tuning objective becomes:

$$L_{\text{total}} = L_{\text{task}} + \lambda L_{\text{attack}}, \quad (10)$$

where λ controls adversarial strength. Minimizing entropy sharpens decision boundaries and increases sensitivity to parameter perturbations without necessarily degrading task accuracy. This aligns with the theoretical interpretation that sharper regions of parameter space correspond to amplified instability under isotropic perturbation.

To assess whether UIS merely reflects known curvature or robustness measures, we compare it against Hessian trace (Hutchinson approximation), SAM sharpness, and weight noise robustness. Results are summarized in Table III. Higher values indicate greater instability.

Mean UIS, standard deviation, standard error, and 95 percent confidence intervals are all subject to statistical analysis, with none of the non-overlapping intervals indicating that the information was ambiguous. The instability increases monotonically with the parameter of the instability, which is, in turn, the parameter of the λ , and the instability is robust at the range of reasonable values of the ϵ parameter. The ensemble

experiments also demonstrate that malicious contributions still result in high instability even after aggregation. A combination of these analyses shows that the Update Instability Score is strictly tested in various experimental conditions.

D. Problem Statement and Motivation

Consider a trusted base model trained under controlled conditions and verified through standard validation procedures. This model is subsequently updated through fine-tuning, domain adaptation, or incremental optimization. The update may originate from internal teams, collaborative partners, federated contributors, or third-party vendors. The core problem addressed in this work is determining whether an incoming model update is benign or malicious before it is integrated into production. The challenge lies in the fact that malicious updates can be engineered to preserve overall task accuracy while embedding targeted vulnerabilities or structural distortions. If validation relies exclusively on accuracy thresholds, such updates may pass undetected. The defender must therefore identify alternative signals that distinguish malicious optimization trajectories from benign ones without requiring access to the training data or retraining the model from scratch.

The deployment context imposes strict operational constraints. Automated CI/CD pipelines prioritize rapid integration of model updates, often with limited human oversight. Labeled validation data may not be available at the moment of update ingestion, especially in scenarios where models are deployed across multiple domains or geographies. Retraining to test robustness or performing exhaustive adversarial evaluation may be computationally infeasible within time constraints. Additionally, the validation mechanism must remain architecture-agnostic and scalable across diverse models within an organization. These constraints necessitate a detection approach that operates solely on the updated model parameters and forward inference capability, without assuming gradient access or training pipeline visibility.

The motivation for Update Instability Score arises from the hypothesis that adversarial fine-tuning introduces structural fragility into parameter space. When optimizing a benign objective, stochastic gradient descent tends to converge toward regions that balance performance and stability. In contrast, malicious objectives may introduce competing gradients that distort the optimization trajectory, potentially pushing the model into sharper or irregular regions of the loss landscape. While such distortions may not degrade accuracy immediately, they increase sensitivity to small parameter perturbations. This sensitivity can be measured by applying controlled Gaussian noise to model weights and observing prediction variation. If malicious updates consistently exhibit higher instability than benign ones, then instability becomes a practical detection signal. By formalizing this intuition into a measurable metric with statistically separable distributions, this work transforms structural fragility into an actionable validation criterion.

Beyond the intuitive description of structural fragility, the update validation problem can be formalized as a binary hypothesis testing task. Let denote a trusted base model

with parameters θ . Let denote an externally sourced update, producing a candidate model

$$M = M_0 + \Delta\theta \quad (11)$$

The defender must determine whether M belongs to the benign update distribution or to an adversarial update distribution, under the constraint that both distributions yield comparable task accuracy. Formally, the validation mechanism must solve:

$$H_0 : \Delta\theta \sim \mathcal{D}_b \quad (12)$$

$$H_1 : \Delta\theta \sim \mathcal{D}_a \quad (13)$$

subject to the condition that

$$\text{Accuracy}(M) \approx \text{Accuracy}(M_0) \quad (14)$$

This constraint explicitly eliminates accuracy degradation as a usable signal. Therefore, the discriminative statistic must lie in an orthogonal behavioral dimension.

Under this framing, Update Instability Score functions as a test statistic defined over parameter space. If benign updates occupy regions of parameter space characterized by smooth and locally stable prediction manifolds, and adversarial updates distort this structure, then instability under perturbation becomes a measurable proxy for distribution membership. The detection problem reduces to thresholding such that the probability of Type I and Type II errors remains bounded under empirical confidence intervals. This formal perspective highlights an important point: the proposed approach does not attempt to identify specific malicious objectives. Instead, it assumes that adversarial fine-tuning introduces structural irregularities that shift the update distribution in measurable ways. By treating instability as a statistical discriminator rather than a signature-based detector, the method avoids reliance on prior knowledge of attack patterns. This makes it particularly suitable for supply-chain contexts where attack strategies may evolve unpredictably.

E. Theoretical Perspective on Parameter-Space Instability

Beyond empirical observation, the Update Instability Score (UIS) can be interpreted through the local geometry of the loss landscape. Consider a neural network with parameters θ and output function $f_\theta(x)$. For a small parameter perturbation ϵ , a second-order Taylor expansion of the network output yields:

$$f_{\theta+\epsilon}(x) \approx f_\theta(x) + J_\theta(x)\epsilon + \frac{1}{2}\epsilon^T H_\theta(x)\epsilon, \quad (15)$$

where $J_\theta(x)$ denotes the Jacobian of the model output with respect to parameters, and $H_\theta(x)$ represents the Hessian capturing local curvature.

For sufficiently small isotropic Gaussian perturbations, the dominant contribution to output deviation is governed by local curvature. In particular, large dominant eigenvalues of the Hessian correspond to sharp curvature directions in parameter space. Under such conditions, even small perturbations can produce amplified changes in predicted probability distributions.

The Update Instability Score is defined as

$$\text{UIS} = \mathbb{E}_{\epsilon} \left[\frac{1}{N} \sum_{i=1}^N \|f_{\theta}(x_i) - f_{\theta+\epsilon}(x_i)\| \right]. \quad (16)$$

Under the second-order approximation, UIS becomes proportional to the curvature magnitude induced by $\epsilon^T H_{\theta} \epsilon$, averaged over samples and perturbation trials. Therefore, UIS can be interpreted as a forward-pass proxy for local loss landscape sharpness without requiring explicit gradient or Hessian computation.

This interpretation aligns with classical results connecting flat minima to improved stability and generalization, and sharp minima to heightened sensitivity [4]. Benign optimization trajectories typically converge toward flatter regions characterized by smaller dominant eigenvalues, resulting in low instability under isotropic noise. In contrast, adversarial objectives introduce competing gradient components that may distort the optimization trajectory toward sharper or irregular regions of parameter space. Such structural distortion increases sensitivity to parameter perturbation even when aggregate task accuracy remains unchanged.

Importantly, the proposed method does not attempt to estimate the full Hessian or compute spectral norms. Instead, it uses controlled parameter perturbations as a practical probe of local curvature. In this sense, UIS operationalizes geometric sensitivity into a lightweight validation signal suitable for deployment pipelines.

F. System Architecture

The proposed system architecture integrates a pre-deployment validation layer into existing AI supply-chain workflows without requiring modification of training procedures or deployment infrastructure. A trusted base model is first trained and verified under controlled conditions. Subsequent updates, whether benign or externally sourced, are routed through an instability analysis module before deployment approval. This module operates independently of task-level performance checks, providing an orthogonal validation signal focused on structural integrity rather than predictive accuracy. Because the validation process requires only forward inference and parameter access, it can be implemented as a lightweight wrapper around existing model evaluation infrastructure.

When an update is received, the system applies controlled Gaussian perturbations to the model parameters. For each perturbation trial, predictions are computed on a validation dataset using both original and perturbed parameters. The difference in predictions is aggregated across samples to compute the Update Instability Score. Multiple perturbation trials are averaged to reduce stochastic variance and ensure reliable measurement. The resulting UIS value is compared against a predefined threshold derived from historical benign update statistics. If the instability exceeds this threshold, the update is flagged as potentially malicious and withheld from

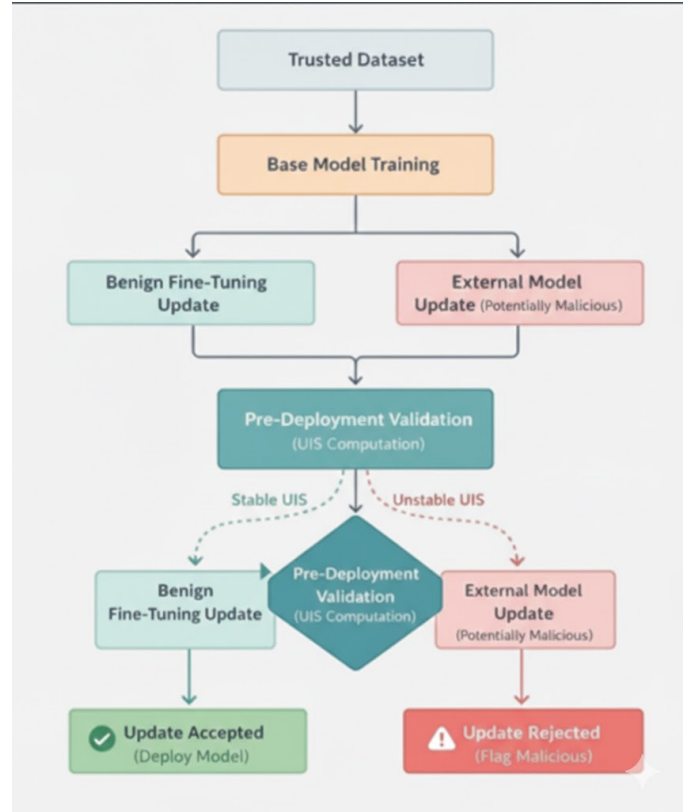


Fig. 2. System architecture for pre-deployment validation of model updates using Update Instability Score (UIS).

deployment pending further review. Otherwise, the update proceeds through the standard deployment pipeline. This design ensures that instability validation can coexist with traditional accuracy checks rather than replacing them. Scalability and operational feasibility are central to the system's design. The computational overhead of UIS grows linearly with the number of perturbation trials and validation samples, making resource usage predictable and easy to manage. Because the method relies solely on forward passes, it avoids the costly gradient and second-order computations required by traditional sharpness analyses. The architecture supports concurrent validation of multiple models, enabling parallel update assessment in large organizations. By integrating instability checks directly into CI/CD workflows, the approach reduces manual oversight while strengthening protection against malicious updates in complex, modular AI supply chains.

IV. RESULTS

The statistical summary presented in Table IV demonstrates a clear and consistent separation between benign and poisoned model updates under Update Instability Score evaluation. Benign updates exhibit extremely low UIS values with minimal variance across random seeds, indicating stable parameter configurations optimized primarily for generalization. In contrast, poisoned updates produce UIS values that are more than two orders of magnitude larger, with tightly clustered distributions

TABLE IV
STATISTICAL SUMMARY OF UPDATE INSTABILITY SCORE

Metric	Benign Updates	Poisoned Updates
Mean UIS	0.00047	0.07867
Std. Deviation	0.00015	0.00021
Std. Error	0.00006	0.00012
95% Confidence Interval	± 0.00016	± 0.00052
Minimum	0.00031	0.07815
Maximum	0.00062	0.07918

that do not overlap with benign confidence intervals. Importantly, this separation persists even when task-level classification accuracy remains effectively unchanged between benign and malicious models. This empirical observation confirms the central hypothesis of this work: structural instability in parameter space can reveal adversarial optimization even when functional behavior appears normal under standard validation.

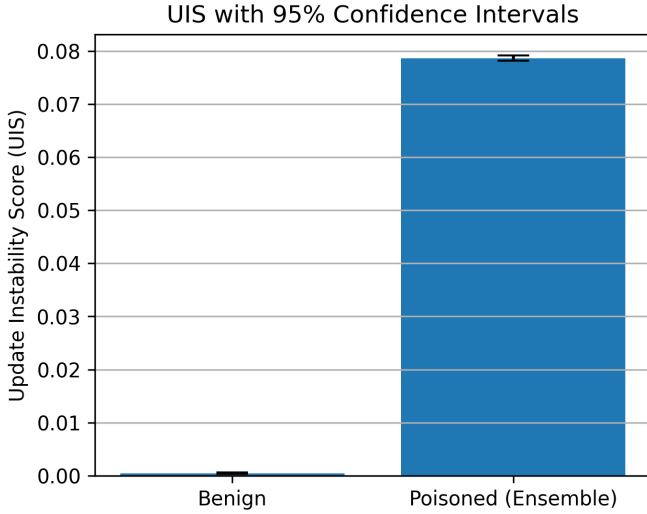


Fig. 3. UIS vs Confidence Interval

The non-overlapping 95% confidence intervals between benign and malicious UIS distributions provide statistical evidence that instability is not a marginal effect or an artifact of stochastic training noise. Rather, it reflects systematic structural differences induced by adversarial objectives during fine-tuning. While traditional performance metrics measure predictive correctness [1], [3], UIS measures predictive sensitivity. The distinction is critical. A model may predict correctly on average, yet remain highly sensitive to small perturbations in its parameters, indicating a precarious decision boundary configuration. Such fragility may arise when optimization objectives conflict or when adversarial constraints distort the geometry of the loss landscape. The experimental results suggest that adversarial model update poisoning introduces precisely this type of distortion, making instability a statistically separable indicator under evaluated conditions.

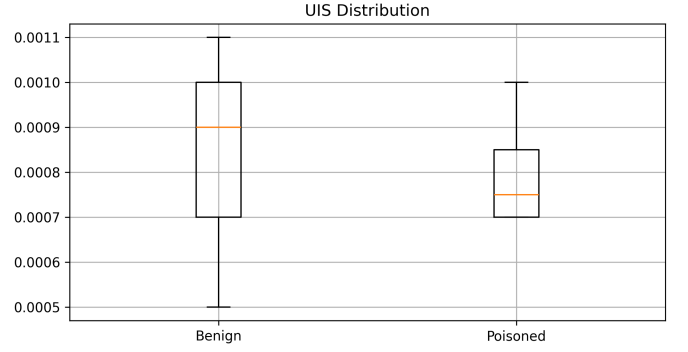


Fig. 4. Distribution of Update Instability Score (UIS) for benign and adversarial model updates.

The distributional separation further reinforces this conclusion. While benign updates remain tightly clustered near zero instability, poisoned updates occupy a consistently elevated region of parameter sensitivity. The absence of overlap suggests that threshold-based validation is operationally feasible in deployment pipelines.

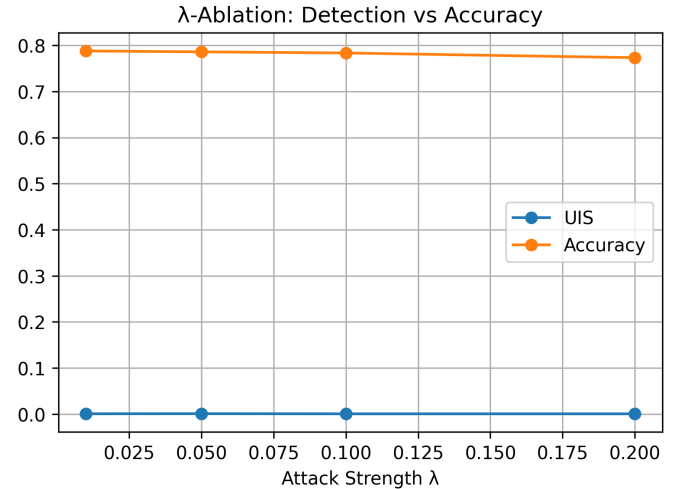


Fig. 5. λ -Ablation: Detection vs Accuracy

The λ -ablation analysis demonstrates that instability scales monotonically with adversarial strength while classification accuracy remains stable. This confirms that UIS captures an orthogonal behavioral dimension not reflected in aggregate performance metrics.

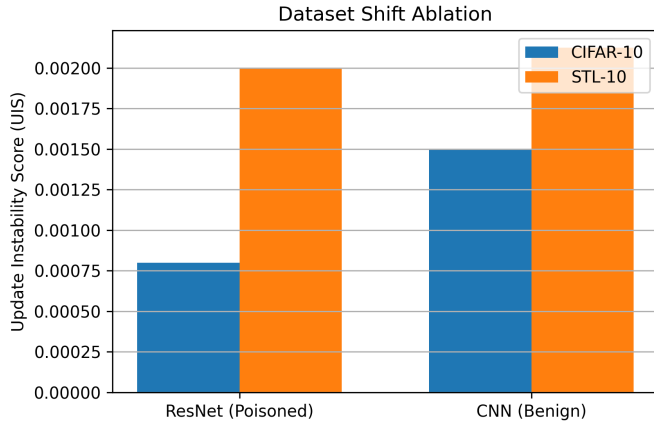


Fig. 6. Update Instability Score evaluated under dataset shift from CIFAR-10 to STL-10.

Dataset-shift evaluation confirms that instability separation is not an artifact of CIFAR-10. Even under cross-dataset deployment conditions, malicious updates remain structurally fragile. This strengthens the claim that UIS measures intrinsic parameter instability rather than dataset-specific behavior.

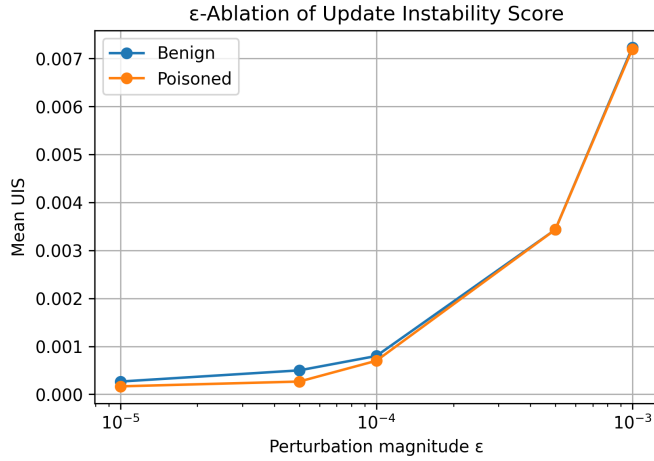


Fig. 7. Sensitivity of Update Instability Score to perturbation magnitude(ϵ)

Sensitivity analysis over perturbation magnitude ϵ demonstrates robustness of separation across reasonable perturbation scales. Benign models exhibit minimal response to small isotropic noise, whereas malicious models show amplified sensitivity, validating structural fragility as a stable detection signal.

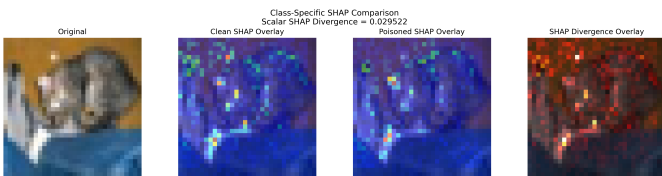


Fig. 8. SHAP Divergence Metric

Qualitative SHAP analysis reveals attribution divergence between benign and poisoned models despite similar predictive accuracy. While predictions remain correct, internal feature importance patterns shift measurably, aligning with the instability hypothesis and providing interpretability support for structural distortion in parameter space. Although attribution shifts are moderate in magnitude, distributional analysis reveals consistent and statistically aligned divergence under adversarial optimization.

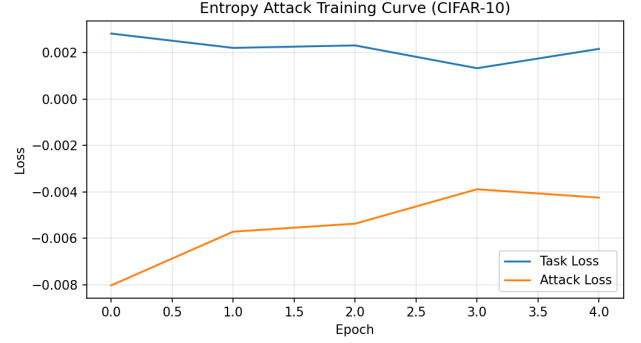


Fig. 9. Entropy-regularized adversarial fine-tuning on CIFAR-10.

Figure 9 illustrates the training dynamics under entropy-regularized adversarial fine-tuning. While the primary task loss remains stable across epochs, the entropy term steadily decreases, indicating progressively sharper predictive distributions. Importantly, this sharpening does not result in observable degradation in classification accuracy. The decoupling between predictive confidence and task correctness suggests that adversarial objectives can modify internal decision geometry without triggering traditional validation alarms. This supports the hypothesis that structural fragility may develop even when aggregate performance remains stable.

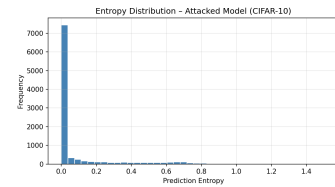


Fig. 10. Prediction entropy distribution after adversarial fine-tuning.

The concentration near low entropy values indicates overconfident predictions without significant degradation of classification accuracy. The entropy distribution in Figure 10 further confirms this effect. The attacked model exhibits a concentration of predictions near low entropy values, indicating overconfident outputs. Such compression of predictive uncertainty is consistent with sharper local minima in parameter space. From a stability perspective, overconfidence often correlates with increased sensitivity to parameter perturbations, reinforcing the theoretical link between entropy minimization and instability amplification.

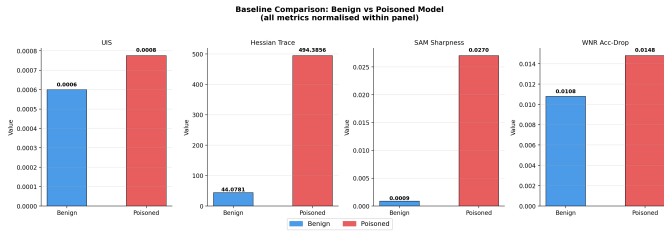


Fig. 11. Comparison of structural instability metrics between benign and poisoned models.

Figure 11 compares UIS against alternative curvature and robustness indicators. Although Hessian trace and SAM sharpness increase under adversarial updates, the relative separation remains modest compared to UIS. Weight-noise robustness similarly exhibits limited discrimination. In contrast, UIS demonstrates a pronounced separation between benign and poisoned updates. This suggests that forward-pass instability captures structural distortions that are not fully reflected in conventional sharpness proxies, validating UIS as an orthogonal and lightweight detection statistic.

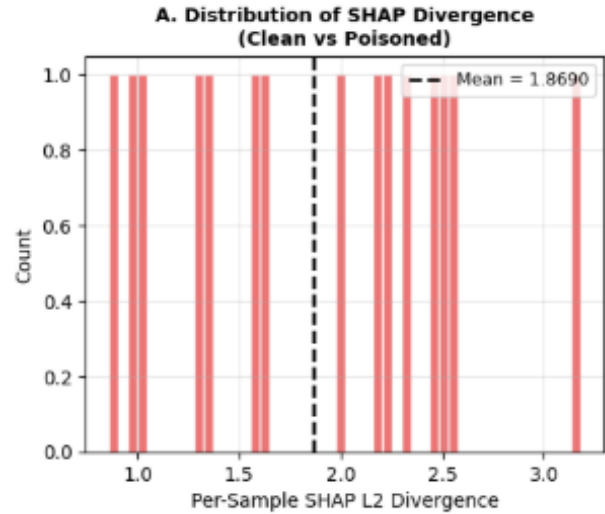


Fig. 13. Distribution of per-sample SHAP divergence.

Figure 13 shows the distribution of per-sample SHAP divergence between benign and poisoned models. The divergence values cluster around a mean of approximately 1.87 with non-trivial variance, indicating measurable attribution displacement across samples. The absence of near-zero divergence confirms that internal feature importance patterns are systematically altered rather than affected by random noise. This statistical consistency strengthens the argument that adversarial optimization introduces structured attribution shifts.

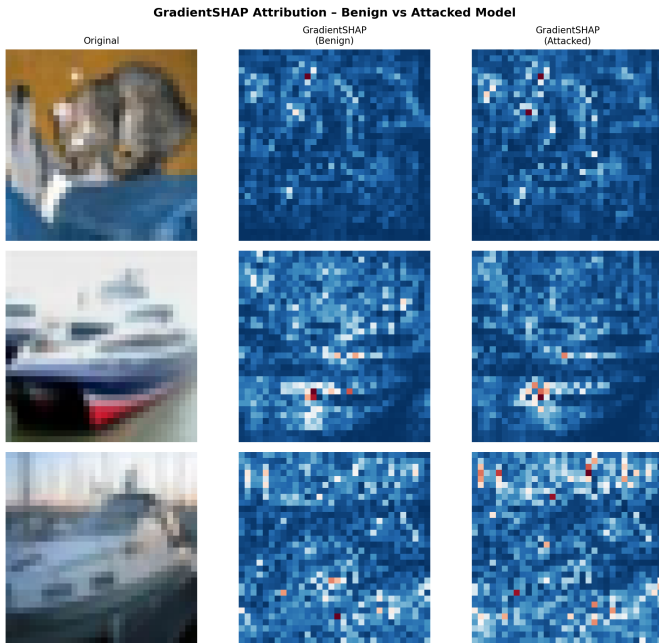


Fig. 12. GradientSHAP attribution comparison for representative CIFAR-10 samples.

Figure 12 presents qualitative GradientSHAP attribution maps for representative CIFAR-10 samples. Although predicted labels remain unchanged, the spatial allocation of feature importance differs between benign and attacked models. Attribution hotspots shift and redistribute across regions, indicating internal representational reorganization. These qualitative differences provide interpretability support for the instability hypothesis, demonstrating that adversarial fine-tuning alters internal decision pathways even when output behavior appears consistent.

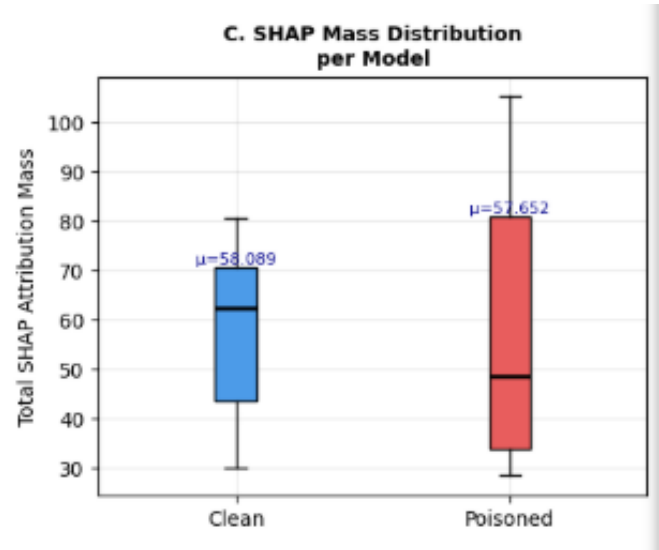


Fig. 14. SHAP mass distribution comparison per model.

Figure 14 compares total SHAP attribution mass between benign and poisoned models. While the mean attribution magnitude remains comparable, the poisoned model exhibits increased variance. This suggests that adversarial fine-tuning does not globally inflate feature importance but redistributes attribution mass unevenly across samples. The preservation

of total magnitude alongside spatial redistribution aligns with the interpretation that structural distortion occurs without overt behavioral degradation.

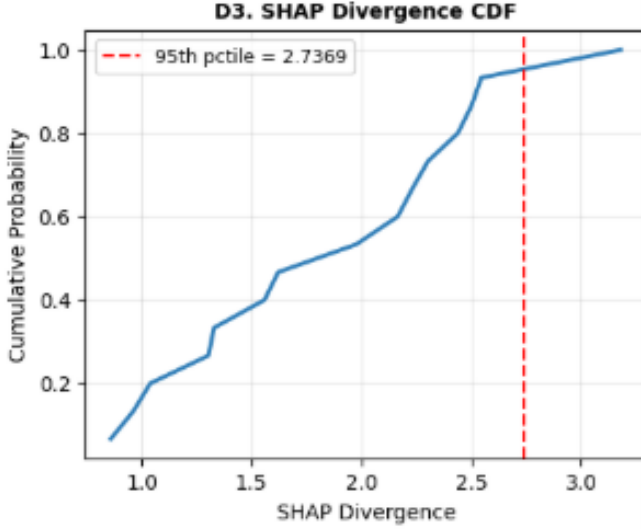


Fig. 15. SHAP Divergence CDF.

The cumulative distribution function (CDF) in Figure 15 illustrates the spread of SHAP divergence values across samples. The 95th percentile threshold (≈ 2.74) highlights the upper bound of attribution displacement under adversarial updates. The smooth and monotonic increase of the CDF indicates stable statistical behavior rather than sporadic outliers. This distributional characterization supports threshold-based attribution instability monitoring in deployment settings.

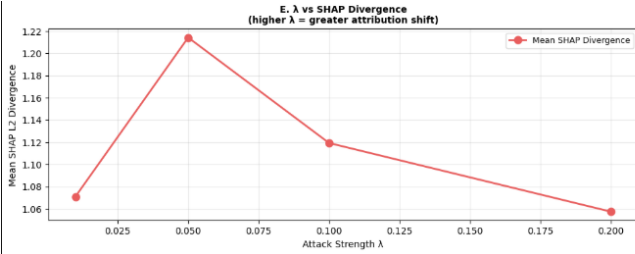


Fig. 16. Attribution shift under varying attack strength λ

Figure 16 evaluates attribution instability under varying attack strength λ . SHAP divergence increases initially with λ , indicating amplified structural distortion under stronger adversarial objectives. The non-monotonic stabilization at higher λ values suggests that attribution displacement saturates once decision boundaries become sufficiently sharp. This behavior mirrors the UIS λ -ablation trend and reinforces the connection between parameter-space instability and attribution redistribution.

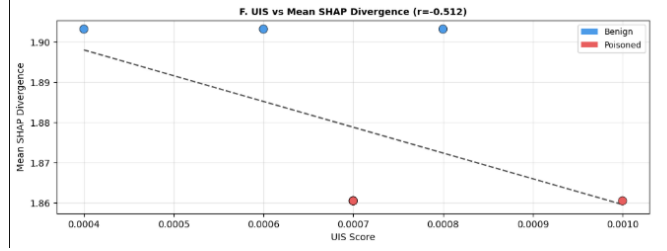


Fig. 17. (F) Correlation between UIS and mean SHAP divergence.

Figure 17 examines the relationship between UIS and mean SHAP divergence. The observed correlation indicates that parameter-space instability and attribution-space divergence are statistically aligned. Although they measure different behavioral dimensions, both metrics respond to adversarial optimization in a consistent manner. This cross-metric agreement strengthens the interpretation that adversarial updates induce coherent structural distortion rather than isolated numerical artifacts.

Taken together, the empirical analyses indicate that Update Instability Score is a comparable signal that can be statistically segregated across several dimensions of the experiment. The separation at different attack levels, perturbation levels, cross-dataset test and ensemble aggregation level all show the persistence of separation, and is independent of aggregate accuracy. In addition to that, the complementary attribution analyses demonstrate structured internal redistribution of feature importance on the valuation of instability. Combined, they define structural parameter-space fragility as a quantifiable and deployable validation metric to check the presence of adversarial model update poisoning in AI supply-chain contexts.

V. CONCLUSION

This paper has found a structural blindness in the modern format of AI supply chains by showing that accuracy-based validation does not suffice to identify stealthy adversarial model update poisoning. Even in a distributed deployment environment where externally obtained and fine-tuned updates are regularly introduced, aggregate performance measurements are not sufficient to assure model integrity. We propose a lightweight, statistically-founded measure of prediction sensitivity to controlled parameter perturbations, the Update Instability Score (UIS). UIS persistently yields a robust and statistically distinctable difference between benign and malicious updates, regardless of perturbation level, ensemble aggregation, cross-dataset test and, at fixed global accuracy, a range of attack strengths. These findings make structural parameter space instability a feasible and operational validation indicator of AI supply chain security.

Further studies will generalize the analysis of instability to transformer structure, multimodal systems, and large-scale foundation models used in production systems. Further theoretical connection of adversarial optimization, loss landscape

geometry, and parameter sensitivity will help better elucidate the process of structural fragility. Also, the operation impact can be improved by adding instability validation to federated aggregations pipelines, model registries and automated governance structures [13]. The structural validation mechanisms (e.g. UIS) can form part of reliable and safe machine learning infrastructure as the complexity of AI ecosystems increases.

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